Solar Energy Forecasting

Predicting daily energy output for a 5 kWp PV system at FH Upper Austria (Hagenberg) using weather data and machine learning.

1. Summary

The primary objective of this project was to apply machine learning techniques to forecast the daily electrical energy output of a rooftop solar photovoltaic (PV) system. Hourly modelled photovoltaic output data (PVGIS-SARAH3, 2005–2023) was integrated with hourly historical weather data from Open-Meteo (ERA5 reanalysis model) to construct a comprehensive daily dataset.

Three baseline models and three machine learning algorithms were trained on historical data (2005–2021) and systematically evaluated on an untouched validation year (2022).

Model / Baseline	Average Miss (MAE)	Model Description / Benchmark Role
Baseline 1: Persistence	6.52 kWh/day	Assumes today's energy output equals yesterday's output.
Baseline 2: Seasonal Average	6.86 kWh/day	Historical 15-day smoothed average output for the specific calendar day.
Baseline 3: Cloud Rule (Best Baseline)	4.75 kWh/day	Theoretical clear-sky max energy reduced by a heuristic cloud cover penalty.
Linear Regression	2.74 kWh/day	Fits a linear mathematical relationship between daily weather features and output.
Random Forest	2.60 kWh/day	Ensemble of decision trees capturing non-linear feature interactions.
Gradient Boosting (Best Model)	2.55 kWh/day	Sequentially boosted decision trees minimizing residual prediction errors.

2. Project Definition

Item / Parameter	Project Specification & Decision
Goal	Predict the total daily energy yield (kWh) of a rooftop solar PV system using weather parameters.
System Setup (Imaginary)	5 kWp peak capacity, 35° tilt, south-facing (0° azimuth), 14% total system losses.
Location	FH3 Building roof, Softwarepark Hagenberg, Upper Austria (Lat: 48.368, Lon: 14.512, Elev: 439 m).
Target Variable	Daily energy output (kWh) derived from PVGIS-SARAH3 hourly modelled satellite data.
Input Features	Daily aggregated weather parameters from Open-Meteo (ERA5): solar irradiance, cloud cover, temperature, wind speed, humidity, and calendar day encoding.
Dataset Period	2005–2023 inclusive (6,939 total days).
Time-Based Split	Train: 2005–2021 (5,959 valid days) | Validation: 2022 (357 valid days) | Test: 2023 (Held out).
Evaluation Metrics & Goals	Primary Metric: Mean Absolute Error (MAE in kWh/day). Secondary: Root Mean Squared Error (RMSE). Primary Goal: Outperform all baseline benchmarks. Secondary Goal: Reduce MAE below 2.00 kWh/day.

3. Columns Definition

3.1 Open-Meteo Weather Variables (Model Inputs)

Column Name	Meaning / Description	Unit / Format
time	Timestamp of the weather observation (UTC)	YYYY-MM-DDTHH:MM:SS
temperature_2m	Ambient air temperature measured at 2 meters above ground	°C
relative_humidity_2m	Atmospheric relative humidity at 2 meters height	%

Column Name	Meaning / Description	Unit / Format
cloud_cover_low	Cloud cover fraction at low altitudes	%
cloud_cover_mid	Cloud cover fraction at mid altitudes	%
cloud_cover_high	Cloud cover fraction at high altitudes	%
wind_speed_10m	Wind speed measured at 10 meters height	m/s
shortwave_radiation	Incoming shortwave solar radiation on a horizontal surface	W/m^2
direct_radiation	Direct beam solar radiation from the sun	W/m^2
diffuse_radiation	Diffuse solar radiation scattered by clouds and atmosphere	W/m^2

3.2 PVGIS Target Dataset Variables

Column Name	Meaning / Description	Unit / Format
Time	Timestamp of the record (Date and UTC Time)	YYYYMMDD:HHMM
P	PV system electrical power output (modeled for 5 kWp system)	Watts (W)
G(i)	Global irradiance on the inclined (slanted) plane of your solar panels	W/m^2
H_sun	Solar elevation angle above the horizon	Degrees (°)
T2m	2-meter ambient air temperature	°C
WS10m	Wind speed measured at 10 meters height	m/s

4.Treatment of the data

1. Missing data stored as zeros

- **Problem:** Some hours in the middle of the day show 0 energy. Example: noon on 17 May 2008. The real sun was about 58° high, but the file said 0.
- **Solution:** Calculate the real sun height. If the sun is more than 3° above the horizon but the file shows 0, mark that hour as missing. Days with such hours (266 days, 3.8% of all days) are not used. That's a small part of the data, so removing them changes little.

2. Different time formats

- **Problem:** The two files write the time differently: 20050101:0010 (PVGIS) and 2005-01-01T00:00 (Open-Meteo).
- **Solution:** Convert both to real dates with pd.to_datetime. Both are in UTC, so the same day lines up in both files.

3. Hourly data, but the models need one value per day

- **Problem:** Both files have one row per hour.
- **Solution:** Combine the 24 hours of each day. Use the **sum** for energy and sunlight, and the **average** for temperature, clouds, wind and humidity.

5. Planned Next Steps

1. Improve the Forecasting Model

2. Publish the Project on GitHub

3. Containerize and Deploy with Docker and Kubernetes

4. Build an LLM Prototype

5. Secure the LLM Prototype